

Lecture 8/Module 1

DC Circuit Analysis and Theorems

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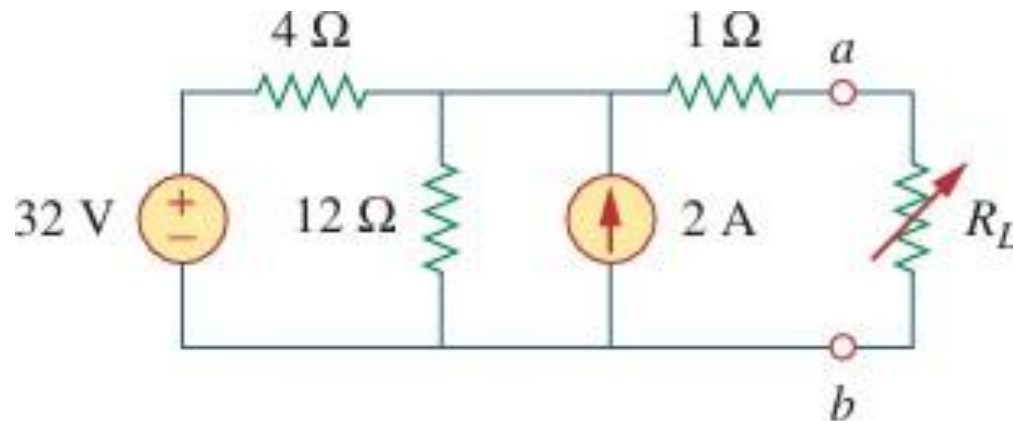


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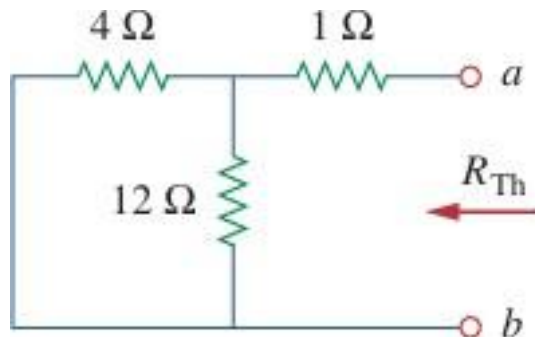
Thevenin's Theorem

Find the Thevenin's equivalent circuit of the circuit shown below, to the left of the terminals.

Then find the current through $R_L = 6\ \Omega$.



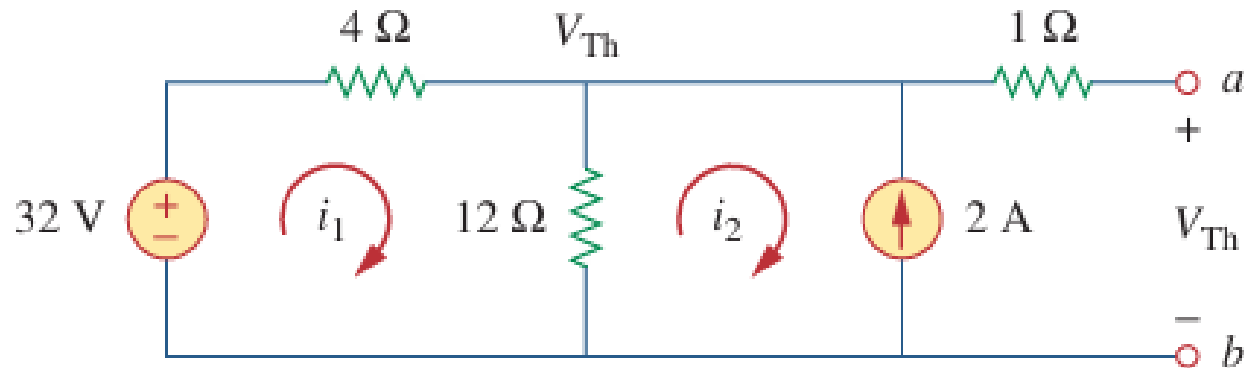
Find R_{th} by removing all the independent sources.



$$R_{Th} = 4 \parallel 12 + 1 = \frac{4 \times 12}{16} + 1 = 4\ \Omega$$

Thevenin's Theorem

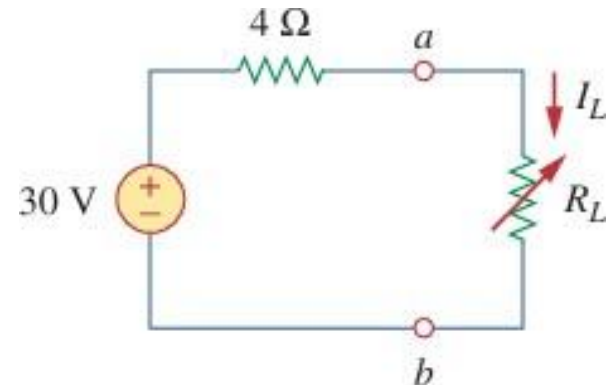
Find V_{th} by removing the load and finding voltage at the terminals a and b.



$$-32 + 4i_1 + 12(i_1 - i_2) = 0, \quad i_2 = -2 \text{ A}$$

$$i_1 = 0.5 \text{ A.}$$

$$V_{Th} = 12(i_1 - i_2) = 12(0.5 + 2.0) = 30 \text{ V}$$



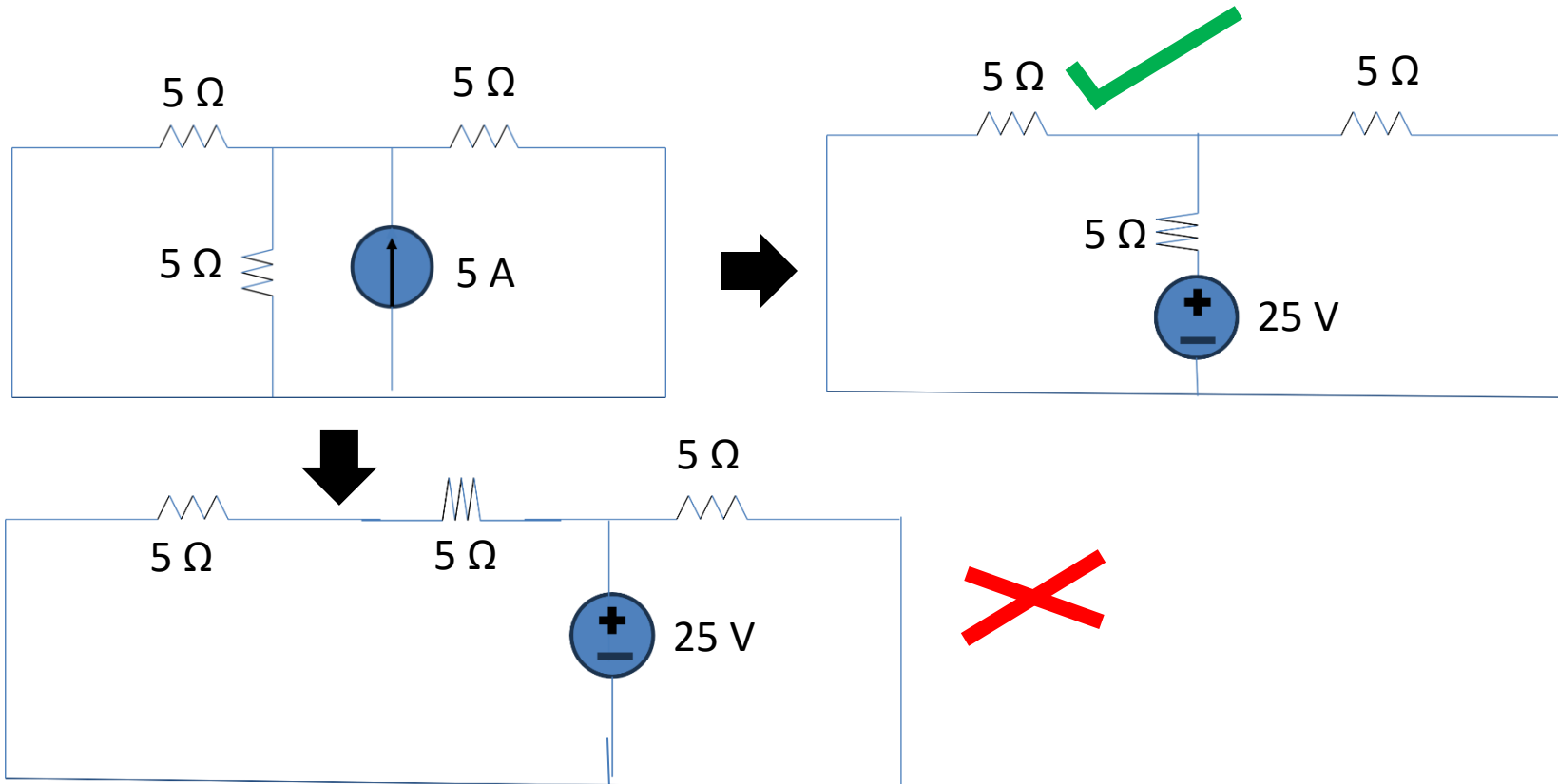
When $R_L = 6$,

$$I_L = \frac{30}{10} = 3 \text{ A}$$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{4 + R_L}$$

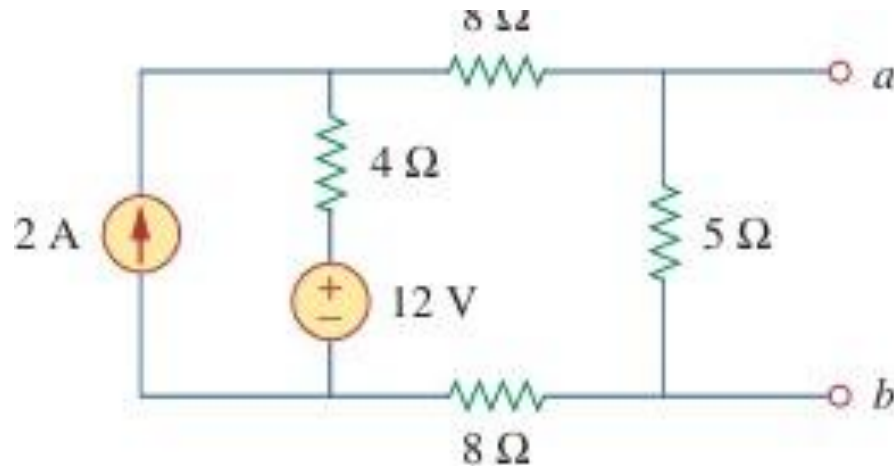
Important Notes

While doing source transformation the end nodes of voltage/current source and series/parallel resistor combined will remain same i.e.; the branches connected to the nodes will be the same.



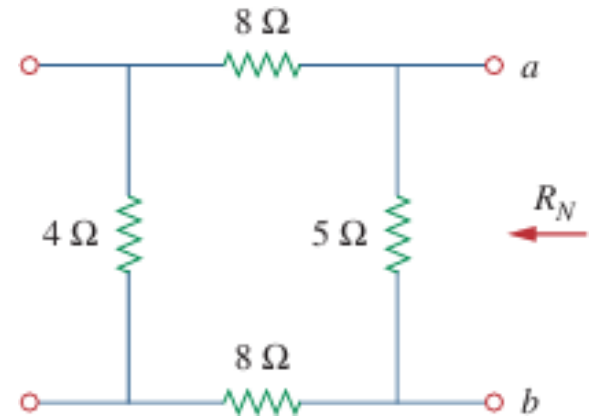
Norton's Theorem

Find the Norton equivalent circuit of the circuit below at terminals a-b.



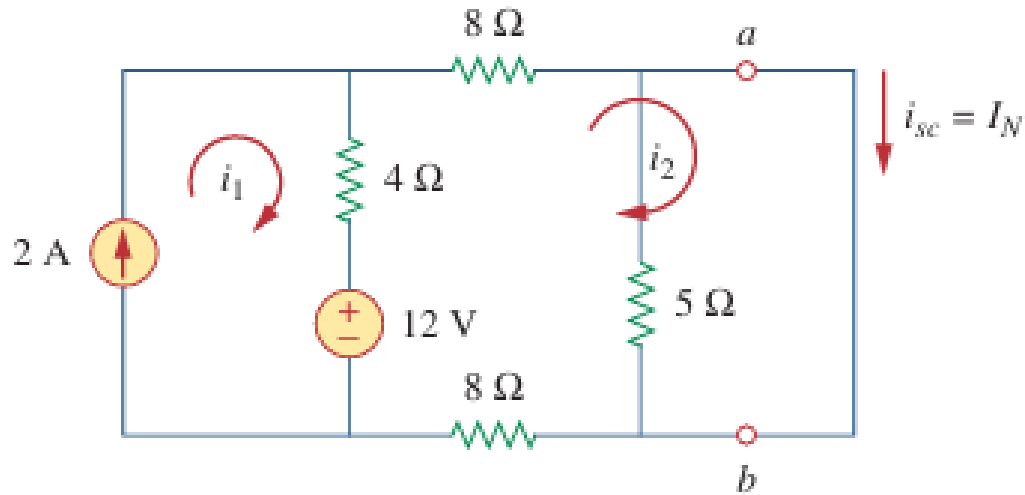
R_N

$$R_N = 5 \parallel (8 + 4 + 8) = 5 \parallel 20 = \frac{20 \times 5}{25} = 4 \Omega$$



Norton's Theorem

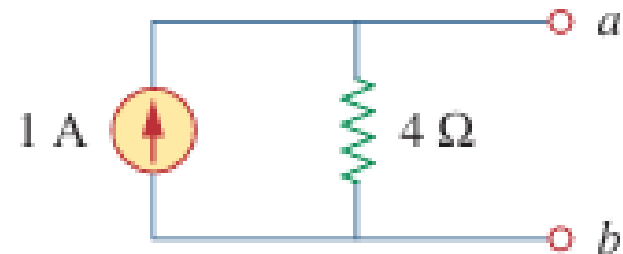
I_N



$$i_1 = 2\text{ A}$$

$$20i_2 - 4i_1 - 12 = 0$$

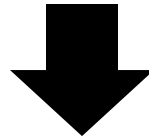
$$i_2 = 1\text{ A} = i_{sc} = I_N$$



Important Notes

One cannot write a mesh equation within a mesh/loop with current source unless the voltage across the current source is known such as node voltages in node analysis.

Relationship between Thevenin's equivalent circuit and Norton's Equivalent Circuit

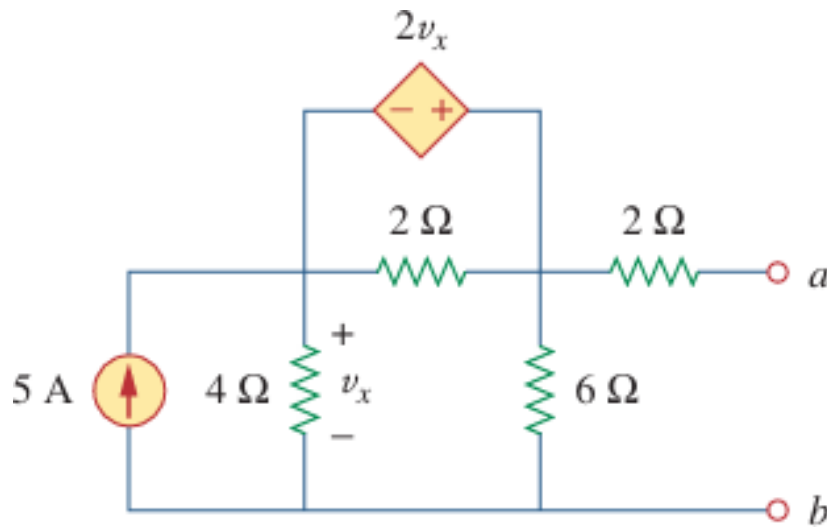


$$R_N = R_{Th}$$

$$I_N = \frac{V_{Th}}{R_{Th}}$$

Thevenin's Theorem with Dependent Source

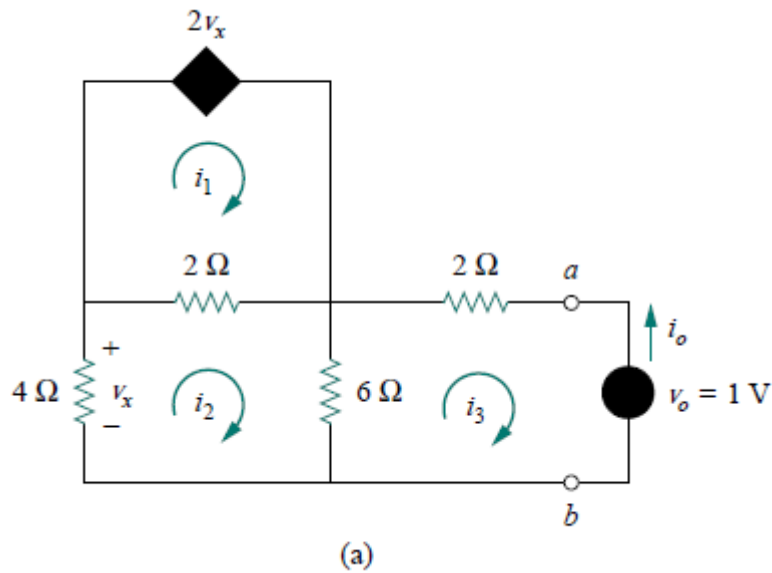
Problem: Find the Thevenin's equivalent of the circuit at terminals a-b.



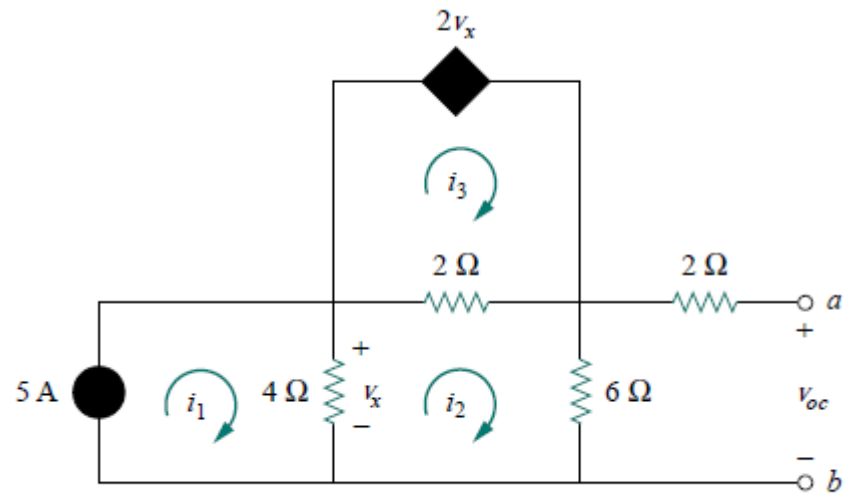
1. To find R_{th} we set the independent source equal to zero but leave the dependent source alone.
2. However, we excite the network with a voltage source v_o connected to the terminals.
3. For easy calculation set $v_o = 1$ V.
4. Let us now find the current i_o through the terminals, and then obtain $R_{th} = v_o/i_o$

Solution: For the given problem, the circuit contains a dependent source. To find R_{Th} , we set the independent source equal to zero but leave the dependent source alone. Because of the presence of the dependent source, however, we excite the network with a voltage source v_0 connected to the terminals as indicated in Fig (a). We may set $v_0 = 1$ V to ease calculation, Since the circuit is linear. Our goal is to find the current i_0 through the terminals, and then Obtain $R_{Th} = 1/ i_0$ (Alternatively we may insert a 1 A current source, find the corresponding Voltage v_0 and obtain $R_{Th} = v_0/1$)

R_{Th}



V_{Th}



Applying mesh analysis to loop 1 in the circuit in Fig. (a) results in

$$-2v_x + 2(i_1 - i_2) = 0 \quad \text{or} \quad v_x = i_1 - i_2$$

But $-4i_2 = v_x = i_1 - i_2$; hence,

$$i_1 = -3i_2$$

For loops 2 and 3, applying KVL produces

$$4i_2 + 2(i_2 - i_1) + 6(i_2 - i_3) = 0$$

$$6(i_3 - i_2) + 2i_3 + 1 = 0$$

Solving these equations gives

$$i_3 = -\frac{1}{6} \text{ A}$$

But $i_o = -i_3 = 1/6 \text{ A}$. Hence,

$$R_{\text{Th}} = \frac{1 \text{ V}}{i_o} = 6 \Omega$$

To get V_{Th} , we find v_{oc} in the circuit of Fig. (b). Applying mesh analysis, we get

$$i_1 = 5$$

$$-2v_x + 2(i_3 - i_2) = 0 \quad \implies \quad v_x = i_3 - i_2$$

$$4(i_2 - i_1) + 2(i_2 - i_3) + 6i_2 = 0$$

or

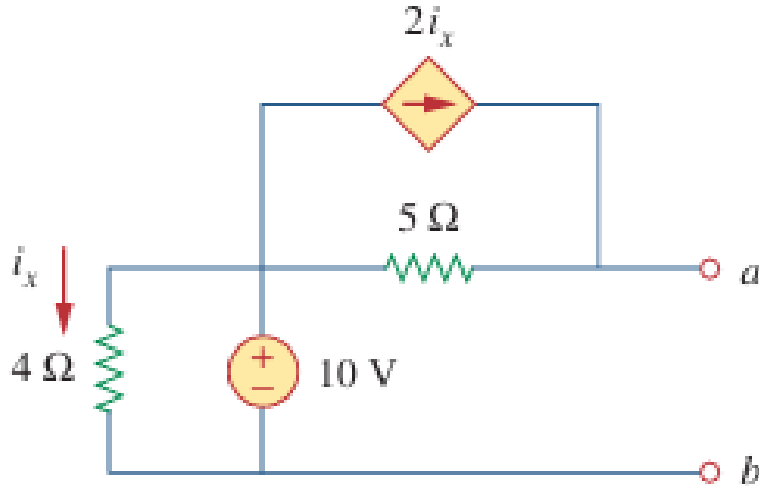
$$12i_2 - 4i_1 - 2i_3 = 0$$

But $4(i_1 - i_2) = v_x$. Solving these equations leads to $i_2 = 10/3$. Hence,

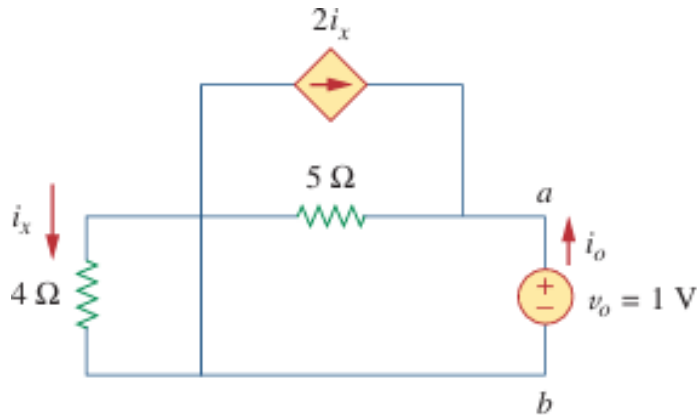
$$V_{\text{Th}} = v_{oc} = 6i_2 = 20 \text{ V}$$

Norton's Theorem with Dependent Source

Find the Norton's equivalent of the circuit at terminals a-b.



R_N



1. To find $R_{th} = R_N$ we set the independent source equal to zero but leave the dependent source alone.

2. However, we excite the network with a voltage source v_o connected to the terminals.

3. For easy calculation set $v_o = 1$ V.

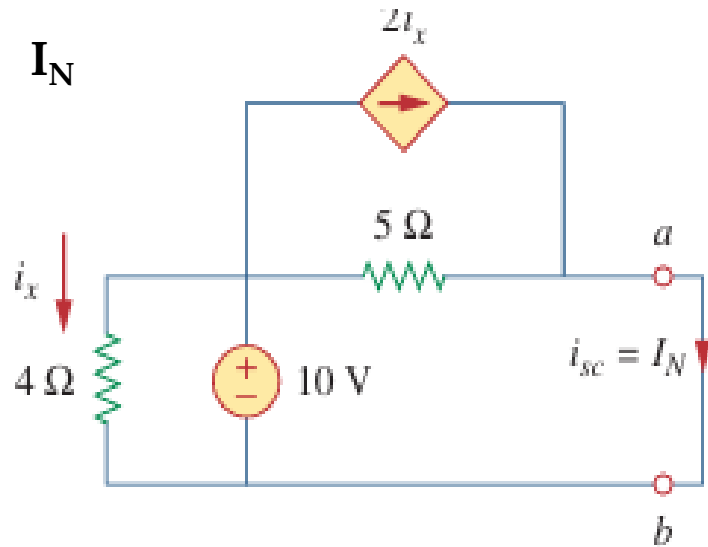
4. Let us now find the current i_o through the terminals, and then obtain $R_N = v_o/i_o$

$$i_x = 0.$$

$$i_o = \frac{1v}{5\Omega} = 0.2 \text{ A}$$

$$R_N = \frac{v_o}{i_o} = \frac{1}{0.2} = 5 \Omega$$

Norton's Theorem with Dependent Source



$4\ \Omega$ resistor, 10 V voltage source, the $5\ \Omega$ resistor, and the dependent current source are all in parallel.

$$i_x = \frac{10}{4} = 2.5\text{ A}$$

At node **a**, KCL gives

$$i_{sc} = \frac{10}{5} + 2i_x = 2 + 2(2.5) = 7\text{ A}$$

$$I_N = 7\text{ A}$$

Maximum Power Transfer Theorem

In practical situations, a circuit is designed to provide power to a load.

There are applications in areas such as communications where it is desirable to maximize the power delivered to a load.

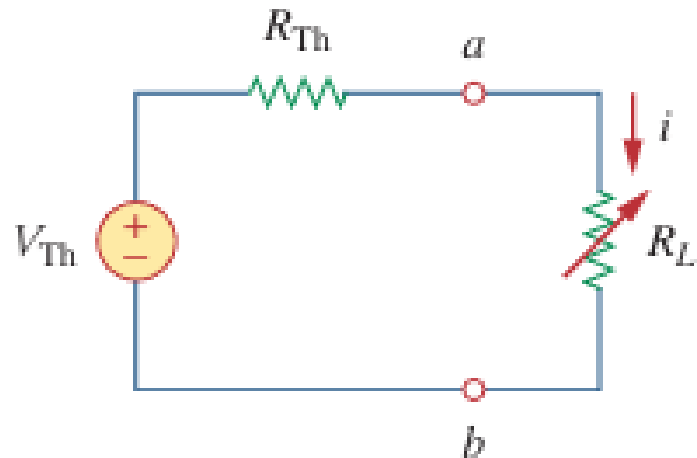
We have to address the problem of delivering the maximum power to a load when given a system with known internal losses.

The Thevenin's equivalent is useful in finding the maximum power a linear circuit can deliver to a load.

We assume that we can adjust the load resistance.

If the entire circuit is replaced by its Thevenin's equivalent except for the load, the power delivered to the load is,

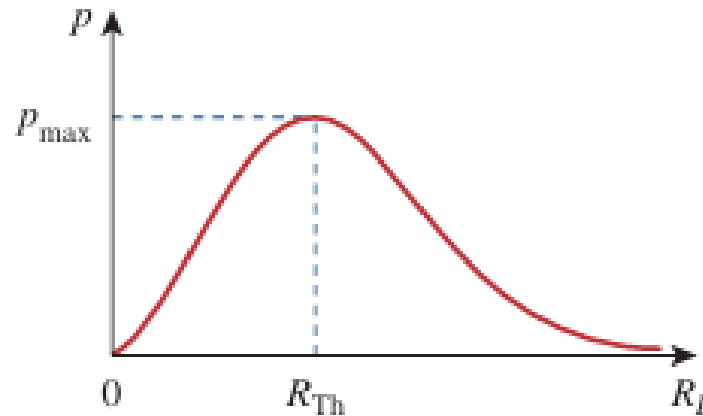
$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$



Maximum Power Transfer Theorem

V_{th} and R_{th} are fixed and by varying the load resistor R_L the power delivered to the load can be varied.

We notice that the power is small for small or large values of R_L but maximum for some value of R_L between 0 and ∞ .



The **maximum power** occurs when $R_L = R_{th}$. This is known as the **maximum power theorem**.

Maximum Power Transfer Theorem

To prove Maximum power transfer theorem, differentiate with respect to R_L and set the result equal to zero:

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L \quad \text{----- 1}$$

$$\frac{dp}{dR_L} = V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right]$$

$$0 = (R_{Th} - R_L)$$

$$R_L = R_{Th}$$

This shows that the maximum power transfer takes place when the load resistance equals the Thevenin resistance ($R_L = R_{th}$)

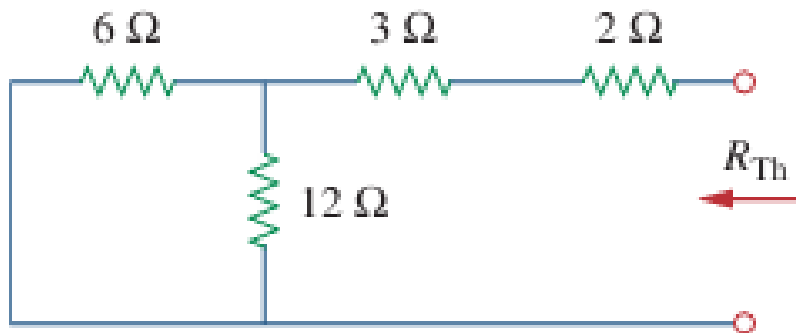
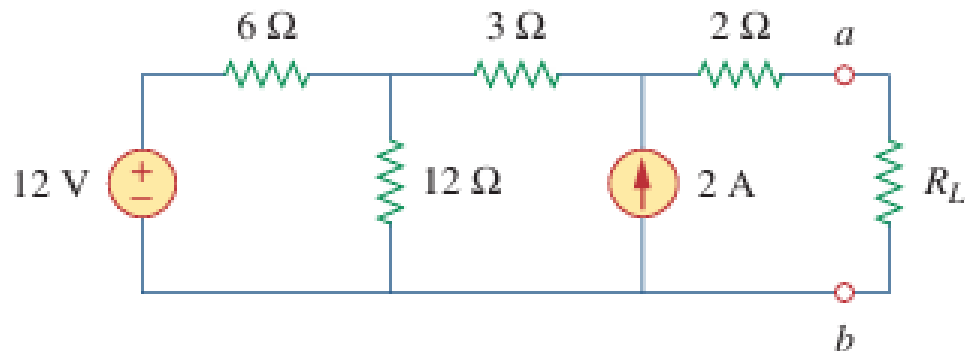
Hence, the maximum power transferred is:

$$p_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

This equation applies only when $R_L = R_{th}$, otherwise we compute the power delivered to the load using equation 1.

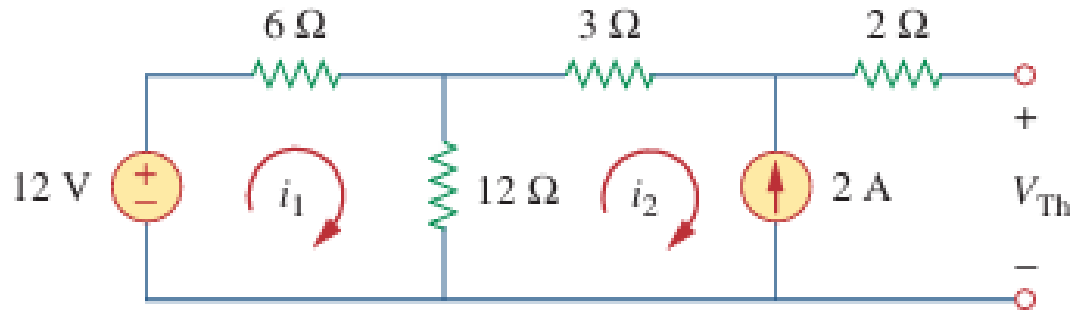
Maximum Power Transfer Theorem

Find the value of R_L for maximum power transfer in the circuit. Find the maximum power.



$$R_{Th} = 2 + 3 + 6 \parallel 12 = 5 + \frac{6 \times 12}{18} = 9 \Omega$$

Maximum Power Transfer Theorem



$$-12 + 18i_1 - 12i_2 = 0, \quad i_2 = -2 \text{ A}$$

$$i_1 = -2/3.$$

$$-12 + 6i_1 + 3i_2 + 2(0) + V_{Th} = 0 \quad \Rightarrow \quad V_{Th} = 22 \text{ V}$$

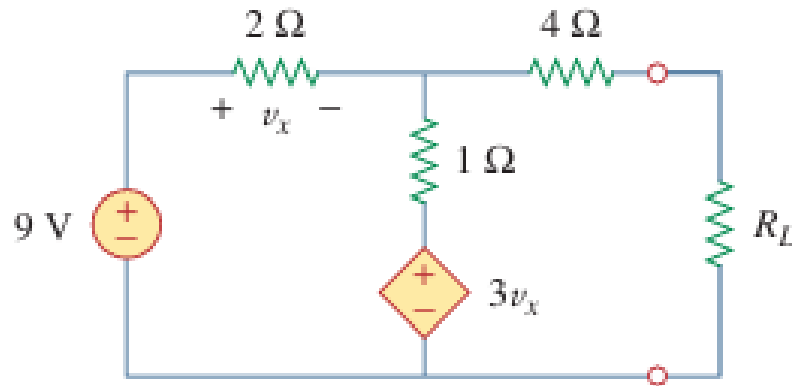
For maximum power transfer $R_L = R_{th}$

$$R_L = R_{Th} = 9 \Omega$$

maximum power is
$$p_{\max} = \frac{V_{Th}^2}{4R_L} = \frac{22^2}{4 \times 9} = 13.44 \text{ W}$$

Maximum Power Transfer Theorem

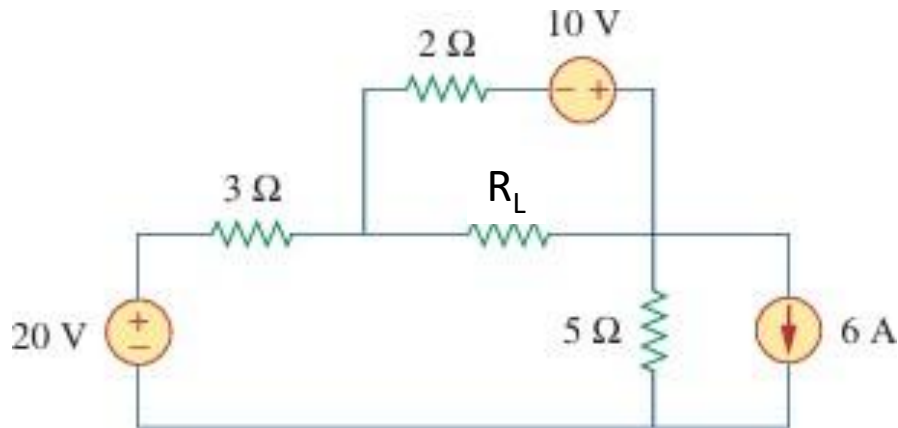
Determine the value of R_L that will draw the maximum power from the rest of the circuit. Calculate the maximum power.



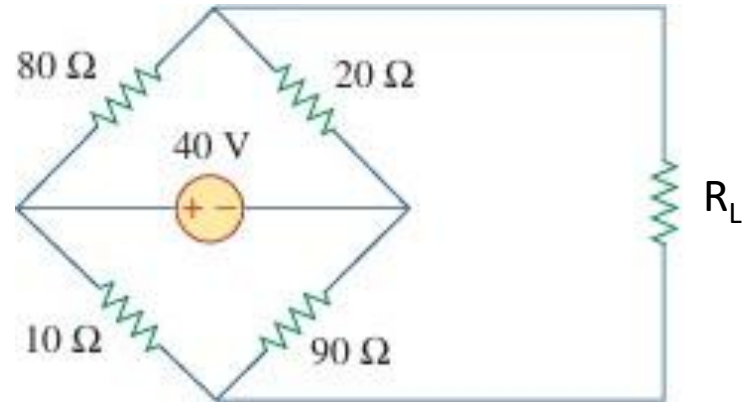
Answer: $R_L = 4.222\ \Omega$, $P_{\max} = 2.901\ \text{W}$

Maximum Power Transfer Theorem

Determine the value of R_L that will draw the maximum power from the rest of the circuit (a) & (b). Calculate the maximum power.



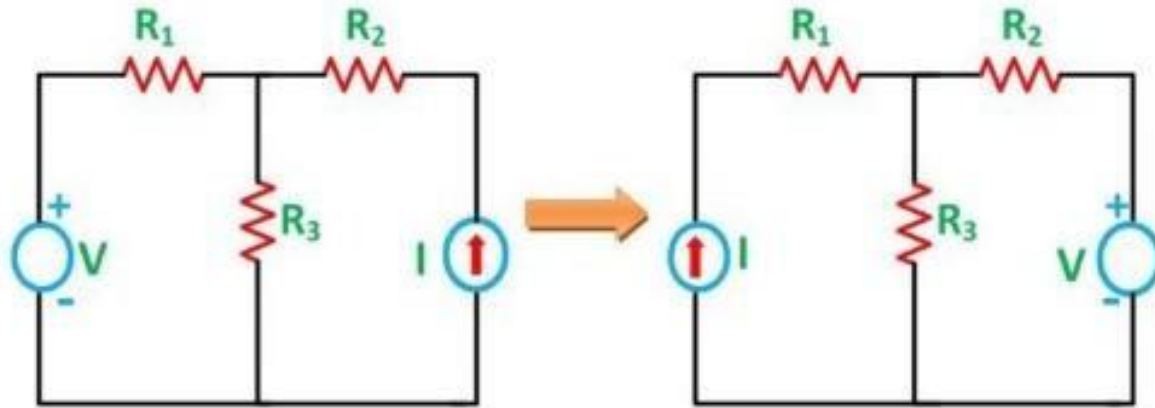
(a)



(b)

Reciprocity Theorem

The reciprocity theorem in electrical circuits states that in a linear, bilateral network with a single voltage source, the ratio of the voltage across one element to the current through another element remains the same even if the voltage source and the current measurement point are interchanged.



Resistances R_1 , R_2 , R_3 are connected in the circuit with a voltage source (V) and a current source (I).

The voltage source and current source can be interchanged for solving the network with the help of Reciprocity Theorem.

Reciprocity Theorem

The limitation of this theorem:

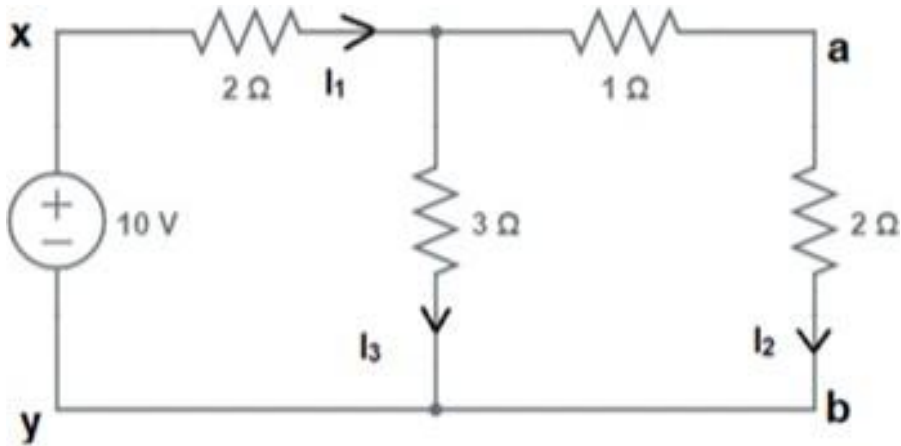
It is applicable only to single-source networks and not in the multi-source network.

The network where reciprocity theorem is applied should be linear and consist of resistors, inductors, capacitors and coupled circuits.

The circuit should not have any time-varying elements.

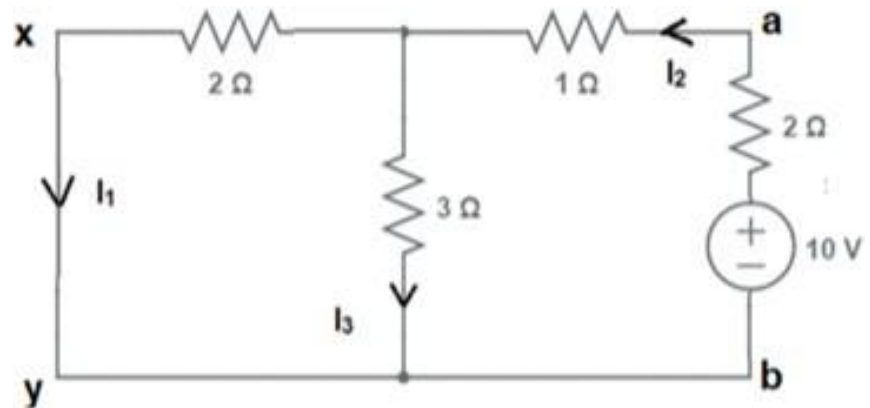
Reciprocity Theorem

Test the validity of the reciprocity theorem in branches x-y and a-b.



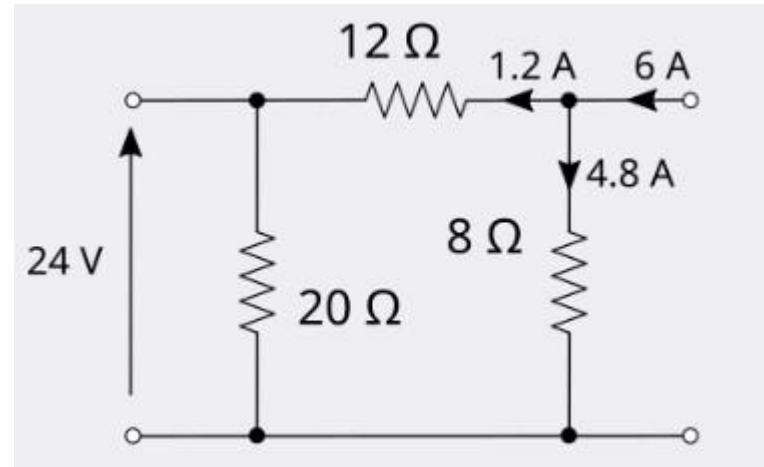
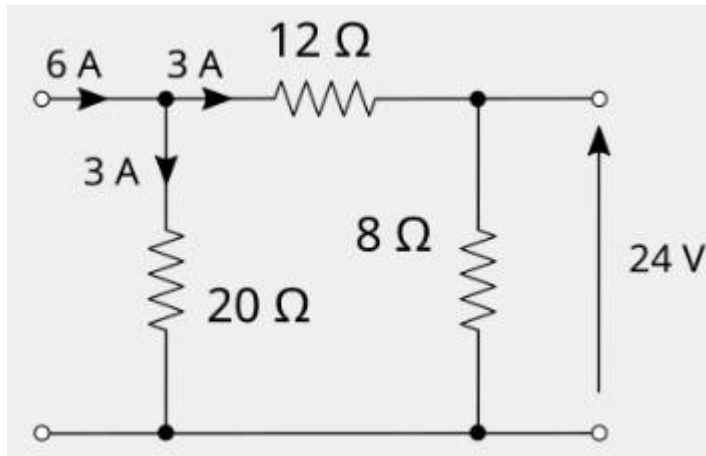
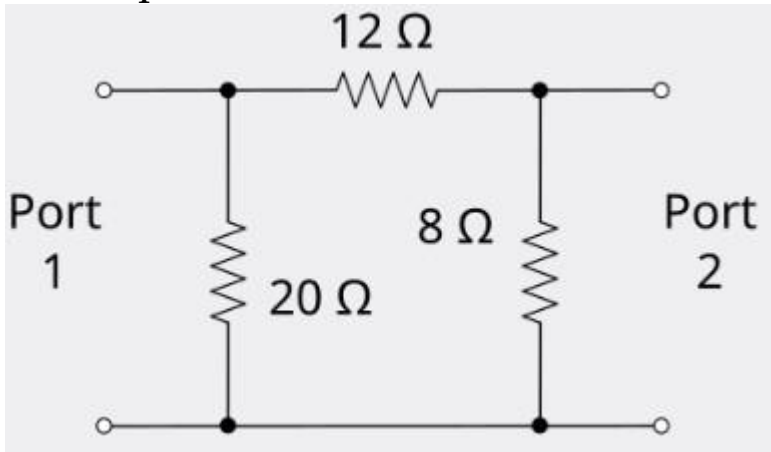
$$I_1 = 1.4 \text{ A}$$

$$I_2 = 1.4 \text{ A}$$



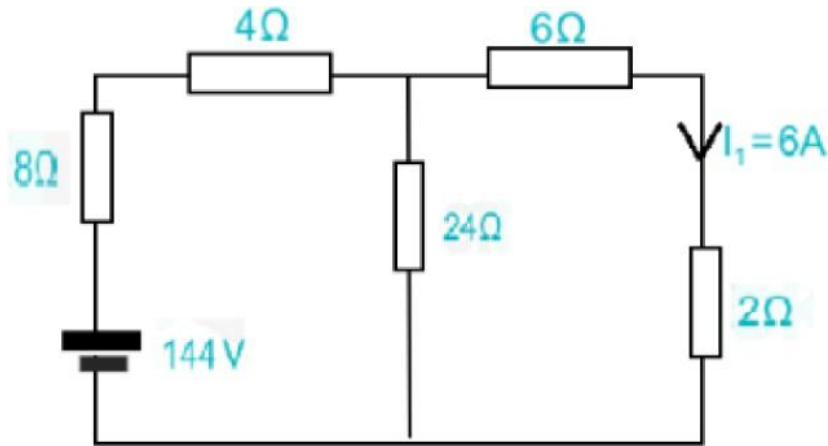
Reciprocity Theorem

Test the validity of the reciprocity theorem in the network if 6A is injected in port 1 and 24 V is measured at port 2.

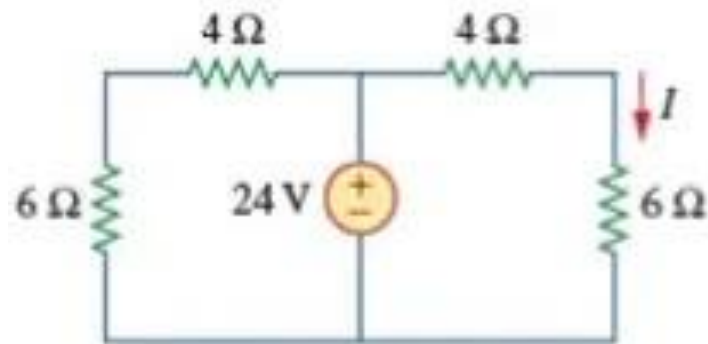


Reciprocity Theorem

Validate the reciprocity theorem.



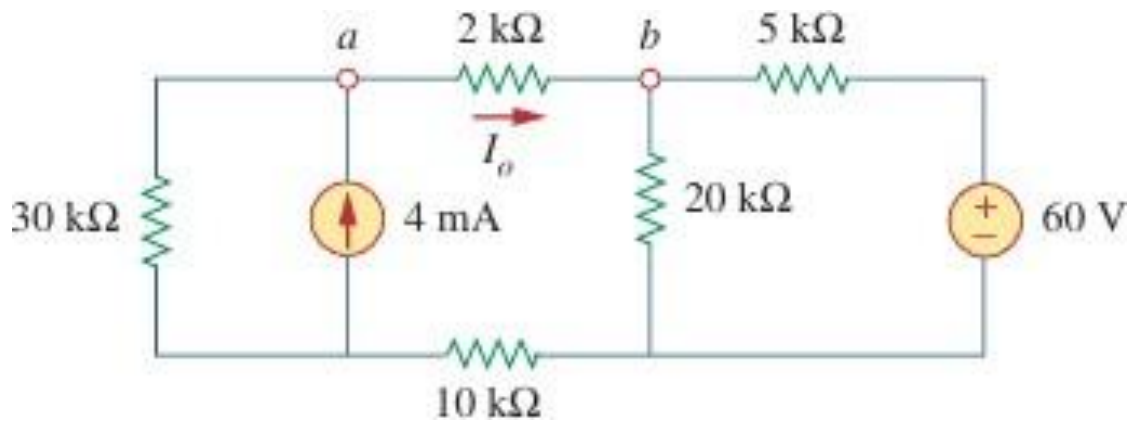
(a)



(b)

Exercise

Consider the circuit given below. An ammeter with internal resistance is inserted between A and B to measure I_o . Determine the reading of the ammeter if: (a) $R_i = 500\ \Omega$ (b) $R_i = 0\ \Omega$ (Hint: Find the Thevenin's equivalent circuit at terminals.



Exercise

Consider the bridge circuit below. Is the bridge balanced? If the 10-k resistor is replaced by an 18-k resistor, what resistor connected between terminals absorbs the maximum power? What is this power?

